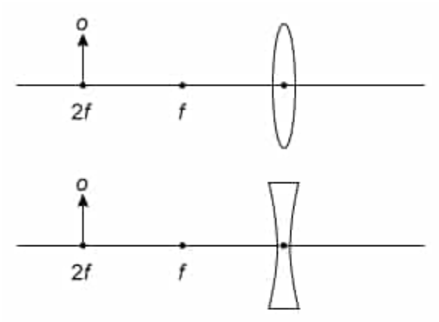
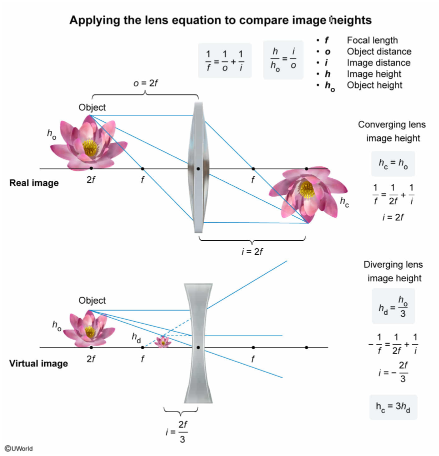


In the figure above, an object *O* is placed a distance 2*f* from the centers of converging and diverging lenses both with focal length *f*. How does the height h_c of the image created by the converging lens compare to the height h_d of the image created by the diverging lens?

- A. $h_c = \frac{1}{3}h_d$ [25%]
- B. $h_c = h_d$ [33%]
- C. $h_c = \frac{3}{2}h_d$ [25%]
- ✔ D. $h_c = 3h_d$ [15%]

Correct
15% answered correctly

Explanation:



The location of the image formed when an object is placed in front of a **thin lens** is determined by the **lens equation**, which states that the lens's focal length *F*, the object distance *o*, and image distance *i* are related:

$$\frac{1}{F} = \frac{1}{o} + \frac{1}{i}$$

Object distance is positive for an object placed in front of the lens. However, the signs of the focal and image lengths depend on whether the lens is a converging lens or diverging lens. The refraction of light through a converging (convex-shaped) lens causes parallel light rays to converge at the focal length behind the lens; hence, *f* is positive. For a diverging (concave-shaped) lens, incident parallel light rays diverge, and *f* is negative.

In this question, *o* = 2*f* for both lenses. In the converging case, *F* = *f*, and:

$$\frac{1}{f} = \frac{1}{2f} + \frac{1}{i}$$

Rearranging and solving for *i* gives:

$$\frac{1}{i} = \frac{1}{f} - \frac{1}{2f}$$

$$\frac{1}{i} = \frac{1}{2f}$$

$$i = 2f$$

In the diverging case, $F = -f$:

$$-\frac{1}{f} = \frac{1}{2f} + \frac{1}{i}$$

Solving for i yields:

$$\frac{1}{i} = -\frac{1}{f} - \frac{1}{2f}$$

$$\frac{1}{i} = -\frac{3}{2f}$$

$$i = -\frac{2f}{3}$$

Geometry implies that the ratio of the image height h and object height h_0 equals the ratio of i and o:

$$\frac{h}{h_o} = \frac{i}{o}$$

Consequently, in the converging case, the image height h_c is:

$$\frac{h_c}{h_o} = \frac{\cancel{2f}}{\cancel{2f}} = 1$$

$$h_c = h_o$$

In the diverging case, the image height h_d equals:

$$\frac{h_d}{h_o} = \frac{\frac{\cancel{2f}}{3}}{\cancel{2f}}$$

$$h_d = \frac{h_0}{3}$$

Therefore, the relation between the heights is:

$$h_c = 3h_d$$

(Choice A) $h_c = \frac{1}{3}h_d$ results from identifying the converging lens as diverging (and vice versa).

(Choice B) Although the object is the same distance from both lenses, the image produced by the converging lens is larger than the virtual image produced by the diverging lens.

(Choice C) $h_c = \frac{3}{2}h_d$ results from missing a factor of 2 in the calculation of the image height in the diverging case.

Educational objective:
The thin lens equation can be used to find the location of an image, given focal length and object distance. The ratio of image height to object height equals the ratio of image distance to object distance.

Time spent:0 sec
Subject:
Physics

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System:
Light and Sound